

RAJKOT, India

WMO: 427370

Lat: 22.309N

Lon: 70.780E

Elev: 441

StdP: 14.46

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	53.7	56.0	24.8	19.2	76.0	29.0	23.4	73.6	17.8	76.6	15.8	76.5	3.1	50	0.410

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	23.9	106.4	71.7	104.4	71.7	102.3	72.4	81.9	90.8	81.0	88.9	80.5	87.8	11.9	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.2	159.4	84.2	79.5	155.8	83.7	78.9	152.7	83.2	46.3	91.2	45.4	88.9	44.7	87.7	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.7	18.6	16.8	DB	49.2	110.1	3.4	1.8	46.8	111.4	44.8	112.5	42.9	113.5	40.5	114.8	
			WB	40.8	84.8	3.3	1.6	38.4	86.0	36.5	86.9	34.6	87.8	32.2	89.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.2	70.0	74.9	82.6	88.7	91.9	89.7	84.3	82.5	83.6	85.4	79.7	72.8
	DBStd	7.35	3.96	4.80	4.32	3.00	2.18	3.49	3.15	2.25	2.27	2.73	3.63	4.11
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	8	5	2	0	0	0	0	0	0	0	0	0	1
	CDD50	11750	621	696	1012	1160	1300	1192	1063	1007	1007	1096	891	705
	CDD65	6285	161	279	547	710	835	742	598	542	557	631	441	242
	CDH74	76604	1437	2753	6569	9807	12221	10357	7012	5732	6187	7839	4570	2120
	CDH80	39760	390	1218	3808	6116	7823	6082	2813	1838	2535	4182	2221	734
Wind	WSAvg	7.8	5.4	6.0	7.1	9.1	11.7	11.2	10.5	9.6	7.8	5.2	4.3	5.1
Precipitation	PrecAvg	25.9	0.0	0.0	0.1	0.0	0.1	4.7	8.2	8.0	3.4	0.4	0.7	0.0
	PrecMax	60.2	0.3	0.3	1.3	0.3	1.0	16.6	18.0	41.9	11.9	1.8	4.5	1.0
	PrecMin	11.7	0.0	0.0	0.0	0.0	0.0	0.0	0.8	0.5	0.0	0.0	0.0	0.0
	PrecStd	11.8	0.1	0.1	0.2	0.1	0.2	4.4	4.4	8.4	3.0	0.5	1.3	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.9	97.0	104.6	107.9	109.8	107.4	97.7	93.3	97.6	101.1	96.4	91.9
		MCWB	64.7	66.2	68.0	69.1	70.9	76.2	78.5	77.9	75.0	71.2	67.8	66.6
	2%	DB	86.9	93.4	101.5	105.4	107.3	103.5	94.7	91.1	94.4	98.8	94.2	89.0
		MCWB	63.0	65.3	66.9	68.8	71.8	77.9	78.5	77.5	76.6	71.2	67.9	65.0
	5%	DB	84.5	90.7	98.7	103.5	105.3	100.8	92.5	89.5	92.3	97.2	92.5	86.8
		MCWB	62.2	64.4	66.5	69.1	72.8	78.5	78.6	77.5	76.6	71.0	67.4	64.0
	10%	DB	81.8	87.5	96.0	101.4	103.0	98.3	90.4	87.8	90.5	95.3	90.6	84.6
		MCWB	61.3	63.5	65.9	69.2	73.5	78.6	78.6	77.6	76.8	71.2	66.8	63.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.3	71.5	74.5	78.2	82.1	83.5	82.6	81.0	81.6	80.6	77.7	73.2
		MCDB	81.4	85.8	91.8	91.5	95.4	93.8	88.3	86.2	88.3	89.9	85.7	85.9
	2%	WB	67.1	69.5	72.4	76.6	80.6	82.3	81.3	80.1	80.5	79.1	75.6	70.0
		MCDB	79.6	82.8	84.8	87.6	91.8	92.7	86.7	85.1	86.9	87.4	83.0	81.7
	5%	WB	65.2	67.9	71.1	75.6	79.7	81.4	80.6	79.4	79.6	78.0	72.9	67.7
		MCDB	78.0	80.8	83.0	85.7	90.3	90.8	85.8	84.2	85.7	86.2	83.4	80.3
	10%	WB	63.4	66.4	69.8	74.7	78.9	80.7	80.0	78.8	78.8	77.1	70.7	65.7
		MCDB	77.0	80.1	82.9	84.8	89.1	89.3	85.2	83.5	84.6	85.2	83.3	79.2
Mean Daily Temperature Range	5% DB	MDBR	23.8	24.5	25.6	26.2	23.9	16.7	10.2	9.7	13.9	20.1	22.1	23.6
		MCDBR	25.4	27.1	28.3	28.1	26.8	21.0	14.0	12.7	17.1	22.7	22.9	24.7
	5% WB	MCWBR	10.2	8.9	7.7	7.8	7.4	4.2	2.9	2.7	3.8	6.8	7.5	10.1
		MCDWR	22.6	23.4	25.0	25.6	22.7	16.7	10.6	9.6	12.3	17.0	18.2	21.3
Clear-Sky Solar Irradiance	taub		0.406	0.418	0.479	0.524	0.568	0.687	0.727	0.656	0.582	0.536	0.522	0.442
	taud		2.182	2.164	2.003	1.934	1.882	1.679	1.624	1.757	1.898	1.921	1.906	2.073
	Ebn at Noon		270	275	263	252	240	211	203	218	233	237	230	252
	Edh at Noon		43	46	56	62	64	78	83	73	62	58	56	46
All-Sky Solar Radiation	RadAvg		1473	1755	2072	2279	2329	1845	1303	1312	1600	1741	1484	1376
	RadStd		44	43	51	40	86	141	111	102	116	70	75	61

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	N/A	N/A	+2.57	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page